D 100613 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER UG (CBCSS-UG) DEGREE EXAMINATION, MARCH 2024

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2020 Admissions onwards)

Time : Two Hours and a Half Maximum Marks : 80

Section A

Answer any number of questions.
Each question carries 2 *marks. Maximum marks* 25.

1. Show that $f(z)=\overline{z}$ is nowhere differentiable.
2. Verify Cauchy-Riemann equations for $f(z) = z^2$.
3. Write the Cauchy-Riemann equations in polar co-ordinates.
4. Define harmonic function and harmonic conjugate function.
5. Solve $e^w = -2$.
6. Express $\cos(2 - 4i)$ in the form $a + ib$.
7. Evaluate $\int ydx + xdy$ on the curve $y = x^2$ from (0, 0) to (1, 1).
8. Evaluate $\oint zdz$ over the first quadrant of the circle $|z| = 1$ from $z = i$ *to* $z = 1$.
9. Prove that $\int_C f(z)dz = 0$ for $f(z)=\frac{z^2}{z-3}$ where C is the unit circle $|z| = 1$.
10. Evaluate $\oint_C \frac{z+1}{z^4+2iz^3}\,dz$ where C is the circle $|z| = 1$.
11. Define (a) Power series ; (b) Circle of convergence.
12. Write the Maclaurin series expansion for $\sin z$ and $\cos z$.
13. Expand $f(z)=e^{\frac{3}{z}}$ in a Laurent series valid for $0<|z|<\infty$.
14. Find zeroes of $f(z)=\frac{z-2}{z^2}\sin\frac{1}{z-1}$.
15. Find the residue of $f(z)=\frac{z}{z^2+1}$ at its poles.

Turn over

Section B

Answer any number of questions.
Each question carries 5 *marks. Maximum marks* 35.

16. Verify the Cauchy Riemann equations for $f(x+iy)=\frac{x-iy}{x^2+y^2}$.

17. If $f(z)=u+iv$ then show that $|f'(z)|^2=u_x^2+v_x^2=u_y^2+u_y^2$.

18. Find the image of the annulus $2\le|z|\le 4$ under the mapping w = Ln z.

19. Find the principal value of $(-3)^{\frac{i}{\pi}}$.

20. Using Cauchy-Goursat theorem, evaluate $\oint_C \frac{5z+7}{z^2+2z-3}dz$ where C is the circle $|z-2|=2$.

21. Find $\oint_C \frac{z^2-4z+4}{z+i}dz$ where C is the circle $|z|=2$.

22. Prove that the sequence $\left\{\frac{3+ni}{n+2ni}\right\}$ converges to $\frac{2}{5}+\frac{1}{5}i$.

23. Find the residue of $f(z)=\tan z$ at $z=\frac{\pi}{2}$.

Section C

Answer any **two** *questions.*
Each question carries 10 *marks. Maximum marks* 20.

24. Verify that the function $u(x,y)=x^3-3xy^2-5y$ is harmonic in the entire complex plane and find the harmonic conjugate function.

25. State and prove (a) Liouville's theorem ; (b) Morerea's thereom.

26. State Cauchy's residue theorem, and using this show that $\int_C \frac{dz}{z\sin z}=0$, where C is the unit circle about the origin described in the positive sense.

27. Show that $\int_0^{2\pi}\frac{d\theta}{2+\cos\theta}\frac{2\pi}{\sqrt{3}}$.

(2 × 10 = 20 marks)